

# High Energy & Particle Physics

## Worked Solutions

BPhO Computational Physics Challenge 2026

### Introduction

This document gives worked solutions to the *High Energy & Particle Physics* problem sheet supplied with the BPhO Computational Physics Challenge 2026.

The calculations and conservation-law checks follow the supplied BPhO solution document. In particular, the particle-physics notation used here follows the sheet: baryon number is  $1/3$  for a quark and  $-1/3$  for an antiquark, lepton number is  $+1$  for leptons and  $-1$  for antileptons, and antiparticles have the same mass but opposite additive quantum numbers.

## 1 Question 1 — Particle interactions and conservation laws

### 1.1 (i) Beta-minus decay

At the quark level,

$$d \rightarrow u + W^-,$$

followed by

$$W^- \rightarrow e^- + \bar{\nu}_e.$$

At the first vertex:

|       | $Q/e$  | $B$   | $L$ |
|-------|--------|-------|-----|
| $d$   | $-1/3$ | $1/3$ | $0$ |
| $u$   | $+2/3$ | $1/3$ | $0$ |
| $W^-$ | $-1$   | $0$   | $0$ |

Hence

$$-\frac{1}{3} = \frac{2}{3} - 1, \quad \frac{1}{3} = \frac{1}{3}, \quad 0 = 0 + 0.$$

At the second vertex,

$$W^- \rightarrow e^- + \bar{\nu}_e,$$

so

$$-1 = -1 + 0,$$

and

$$L(W^-) = 0 = 1 + (-1).$$

Thus charge, baryon number and lepton number are all conserved at both vertices.

## 1.2 (ii) Beta-plus decay

The corresponding process is

$$u \rightarrow d + W^+,$$

followed by

$$W^+ \rightarrow e^+ + \nu_e.$$

At the first vertex,

$$+\frac{2}{3} = -\frac{1}{3} + 1, \quad \frac{1}{3} = \frac{1}{3} + 0, \quad 0 = 0 + 0.$$

At the second vertex,

$$+1 = +1 + 0,$$

and

$$0 = (-1) + 1.$$

Therefore

|                                                                        |
|------------------------------------------------------------------------|
| charge, baryon number and lepton number are conserved at every vertex. |
|------------------------------------------------------------------------|

## 1.3 (iii) $\pi^- + p \rightarrow n + \pi^- + \pi^+$

Using

$$\pi^- = d\bar{u}, \quad p = uud, \quad n = udd, \quad \pi^+ = u\bar{d},$$

the reaction is

$$d\bar{u} + uud \rightarrow udd + d\bar{u} + u\bar{d}.$$

For charge,

$$Q_{\text{initial}} = -1 + 1 = 0, \quad Q_{\text{final}} = 0 - 1 + 1 = 0.$$

For baryon number,

$$B_{\text{initial}} = 0 + 1 = 1, \quad B_{\text{final}} = 1 + 0 + 0 = 1.$$

Thus both charge and baryon number are conserved. The reaction written with an incoming  $\pi^+$  would violate charge conservation; the charge-conserving form above is the one used in the supplied solution.

## 1.4 (iv) $K^+ \rightarrow \pi^+ + \pi^0$

The charged kaon has quark content

$$K^+ = u\bar{s}.$$

The charged pion is

$$\pi^+ = u\bar{d},$$

while the neutral pion is a superposition of

$$u\bar{u} \quad \text{and} \quad d\bar{d}.$$

The initial charge is

$$Q(K^+) = +\frac{2}{3} + \frac{1}{3} = +1.$$

The final charge is

$$Q(\pi^+) + Q(\pi^0) = (+1) + 0 = +1.$$

Therefore

|                                               |
|-----------------------------------------------|
| $Q_{\text{initial}} = Q_{\text{final}} = +e.$ |
|-----------------------------------------------|

### 1.5 (v) Quark binding energies

The supplied masses are

$$m_{K^+} = 0.52616m_p, \quad m_{\pi^+} = 0.14875m_p,$$

and

$$m_p c^2 = 938.27 \text{ MeV}.$$

For the kaon,

$$m_{K^+} c^2 = 0.52616(938.27) = 493.68 \text{ MeV}.$$

The constituent quark masses are

$$m_u c^2 = 2.4 \text{ MeV}, \quad m_s c^2 = 95 \text{ MeV}.$$

Define the binding contribution by

$$B = (m_u + m_s - m_{K^+}) c^2.$$

Then

$$B_{K^+} = 2.4 + 95 - 493.68 = -396.28 \text{ MeV}.$$

Hence

$$\boxed{B_{K^+} \approx -396.2 \text{ MeV}}.$$

The negative sign indicates that the hadron mass-energy is much lower than the sum of the quoted bare quark masses in this simplified constituent picture.

For the pion,

$$m_{\pi^+} c^2 = 0.14875(938.27) = 139.57 \text{ MeV}.$$

Therefore

$$B_{\pi^+} = 2.4 + 4.8 - 139.57 = -132.37 \text{ MeV},$$

so

$$\boxed{B_{\pi^+} \approx -132.4 \text{ MeV}}.$$

These should not be interpreted as ordinary potential energies of two free quarks: quarks are confined, and the simple subtraction is only the model requested by the problem.

### 1.6 (vi) $\Sigma^- \rightarrow n + e^- + \bar{\nu}_e$

The  $\Sigma^-$  has quark content

$$\Sigma^- = dds,$$

while

$$n = udd.$$

At quark level, one strange quark changes through the weak interaction:

$$s \rightarrow u + W^-,$$

followed by

$$W^- \rightarrow e^- + \bar{\nu}_e.$$

At the first vertex,

$$\begin{aligned} -\frac{1}{3} &= \frac{2}{3} - 1, \\ \frac{1}{3} &= \frac{1}{3} + 0, \end{aligned}$$

and lepton number remains zero.

At the second vertex,

$$-1 = -1 + 0$$

and

$$0 = 1 - 1.$$

Thus all three relevant additive conservation laws hold at each vertex.

## 1.7 (vii) Cloud and bubble chambers

A **cloud chamber** contains a supersaturated vapour. A charged high-energy particle passing through the chamber ionises molecules along its path. The ions act as nucleation sites, so droplets condense along the trajectory and form a visible track.

A **bubble chamber** contains a liquid maintained in a superheated state. A charged particle ionises the liquid along its path, providing nucleation sites for bubbles. The bubbles grow along the trajectory and form a visible track.

A magnetic field can be used with either technique to curve charged-particle tracks, allowing the momentum and charge sign to be inferred.

## 1.8 (viii) Pair production of muons

To create a muon–antimuon pair,

$$\gamma \rightarrow \mu^- + \mu^+,$$

the photon must supply at least the two rest-mass energies:

$$E_{\gamma,\min} = 2m_\mu c^2.$$

With

$$m_\mu c^2 = 105.66 \text{ MeV},$$

we obtain

$$E_{\gamma,\min} = 2(105.66) = 211.32 \text{ MeV}.$$

Using

$$1 \text{ GeV} = 1.6 \times 10^{-10} \text{ J},$$

or equivalently

$$1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J},$$

gives

$$E_{\gamma,\min} \approx 211.32(1.6 \times 10^{-13}) = 3.38 \times 10^{-11} \text{ J}.$$

Thus

$$\boxed{E_{\gamma,\min} \approx 3.38 \times 10^{-11} \text{ J}}.$$

The minimum wavelength follows from

$$E = \frac{hc}{\lambda},$$

so

$$\lambda_{\max} = \frac{hc}{E_{\gamma,\min}} \approx 5.88 \times 10^{-15} \text{ m}.$$

Therefore

$$\boxed{\lambda_{\max} \approx 5.9 \text{ fm}}.$$

At threshold the pair is produced with negligible kinetic energy in the idealised threshold calculation.

**1.9 (ix) Electron–positron pair moving at  $0.8c$** 

The total energy of one electron or positron is

$$E = \gamma mc^2,$$

where

$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}}.$$

For

$$u = 0.8c,$$

$$\gamma = \frac{1}{\sqrt{1 - 0.8^2}} = \frac{5}{3}.$$

The photon must supply the total energy of both particles:

$$E_\gamma = 2\gamma m_e c^2.$$

Since

$$m_e c^2 = 0.51100 \text{ MeV},$$

we find

$$E_\gamma = 2 \left( \frac{5}{3} \right) (0.51100) \approx 1.70 \text{ MeV}.$$

Hence

$$\boxed{E_\gamma \approx 1.70 \text{ MeV}}.$$

**1.10 (x) Bremsstrahlung shower**

A high-energy electron entering matter is strongly deflected by the electric fields of atomic nuclei. Because it is accelerating, it emits bremsstrahlung photons.

A sufficiently energetic photon can then undergo pair production:

$$\gamma \rightarrow e^- + e^+.$$

The newly created electron and positron are themselves deflected by nuclei and can emit further bremsstrahlung photons. Those photons can again produce pairs.

Schematically,

$$e^- \longrightarrow e^- + \gamma \longrightarrow e^- + e^- + e^+ \longrightarrow \text{many } e^\pm + \gamma \longrightarrow \text{particle shower}.$$

The repeated branching produces a cascade of particles. The resulting shower can therefore provide an observable signature of the original high-energy electron.

**1.11 (xi) Koide mass ratio for  $u, d, s$** 

The given relation is

$$\frac{m_u + m_d + m_s}{(\sqrt{m_u} + \sqrt{m_d} + \sqrt{m_s})^2} \approx \frac{k}{9}.$$

Using

$$m_u = 2.4, \quad m_d = 4.8, \quad m_s = 95$$

in  $\text{MeV}/c^2$ ,

$$R_q = \frac{2.4 + 4.8 + 95}{(\sqrt{2.4} + \sqrt{4.8} + \sqrt{95})^2}.$$

Numerically,

$$R_q \approx 0.562.$$

Therefore

$$k \approx 9(0.562) \approx 5.06.$$

The requested integer is

$$\boxed{k \approx 5}.$$

## 2 Question 2 — Strangeness

### 2.1 (i) Determining the quark content of $K^0$

The reaction is

$$K^- + p \rightarrow \Omega^- + K^+ + K^0.$$

The  $\Omega^-$  has quark content

$$\Omega^- = sss$$

and hence strangeness

$$S = -3.$$

The initial  $K^-$  has

$$K^- = \bar{u}s,$$

so

$$S_{\text{initial}} = -1.$$

The proton has

$$S = 0.$$

The final  $K^+$  has

$$K^+ = u\bar{s}$$

and therefore

$$S = +1.$$

For strangeness to be conserved, the neutral kaon must supply

$$-1 = (-3) + (+1) + S(K^0),$$

giving

$$S(K^0) = +1.$$

A neutral meson with strangeness +1 must contain an anti-strange quark. The only neutral choice among the light quark flavours is

$$\boxed{K^0 = d\bar{s}},$$

for which

$$Q = -\frac{1}{3} + \frac{1}{3} = 0, \quad S = +1, \quad B = 0.$$

Thus

$$\boxed{K^0 = d\bar{s}}.$$

## 2.2 (ii) $\Omega^- \rightarrow \Xi^0 + \pi^-$

The quark compositions are

$$\Omega^- = sss, \quad \Xi^0 = uss, \quad \pi^- = d\bar{u}.$$

The weak process can be represented by

$$s \rightarrow u + W^-,$$

followed by

$$W^- \rightarrow d + \bar{u}.$$

The first vertex conserves charge:

$$-\frac{1}{3} = \frac{2}{3} - 1,$$

and baryon number:

$$\frac{1}{3} = \frac{1}{3} + 0.$$

The second vertex conserves charge:

$$-1 = -\frac{1}{3} - \frac{2}{3},$$

and baryon number:

$$0 = \frac{1}{3} - \frac{1}{3}.$$

However, strangeness changes:

$$S(\Omega^-) = -3,$$

while

$$S(\Xi^0) = -2, \quad S(\pi^-) = 0.$$

Thus

$$-3 \neq -2.$$

Therefore

$$\boxed{\text{strangeness is not conserved in this weak decay}},$$

as expected for a weak interaction.

## 3 Question 3 — Tau decay

### 3.1 (i) Feynman diagram

The decay

$$\tau^- \rightarrow \pi^- + \nu_\tau$$

occurs through

$$\tau^- \rightarrow W^- + \nu_\tau,$$

followed by

$$W^- \rightarrow d + \bar{u},$$

which hadronise to form the  $\pi^-$ .

Thus, schematically,

$$\tau^- \longrightarrow \nu_\tau + W^-, \quad W^- \longrightarrow d + \bar{u} \longrightarrow \pi^-.$$

### 3.2 (ii) Why is the tau neutrino required?

The tau lepton has lepton number

$$L_\tau = +1.$$

The  $W^-$  boson has

$$L_\tau = 0.$$

Therefore the vertex

$$\tau^- \rightarrow W^- + \nu_\tau$$

requires

$$1 = 0 + 1.$$

Without the tau neutrino, tau-lepton number would change from +1 to 0.

Hence

|                                                               |
|---------------------------------------------------------------|
| tau lepton number would not be conserved without $\nu_\tau$ . |
|---------------------------------------------------------------|

### 3.3 (iii) Kinetic energy of the tau

The tau has

$$m_\tau c^2 = 1776.86 \text{ MeV}$$

and speed

$$u = 0.2c.$$

Its Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1 - 0.2^2}} \approx 1.02062.$$

The total energy is

$$E_\tau = \gamma m_\tau c^2.$$

The kinetic energy is

$$K_\tau = (\gamma - 1)m_\tau c^2.$$

Therefore

$$K_\tau = (1.02062 - 1)(1776.86) \approx 36.64 \text{ MeV}.$$

Thus

|                                     |
|-------------------------------------|
| $K_\tau \approx 36.6 \text{ MeV}$ . |
|-------------------------------------|

### 3.4 (iv) Energy of the tau neutrino

The pion has speed

$$u_\pi = 0.99c.$$

Its mass is given by

$$m_\pi c^2 = 0.14875 m_p c^2 = 0.14875(938.27) \approx 139.57 \text{ MeV}.$$

Its Lorentz factor is

$$\gamma_\pi = \frac{1}{\sqrt{1 - 0.99^2}} \approx 7.0888.$$

Therefore

$$E_\pi = \gamma_\pi m_\pi c^2 \approx 989.37 \text{ MeV}.$$

The tau total energy is

$$E_\tau = \gamma_\tau m_\tau c^2 \approx 1813.50 \text{ MeV}.$$



Assuming the tau neutrino is effectively massless,

$$E_{\nu_\tau} = E_\tau - E_\pi.$$

Hence

$$E_{\nu_\tau} \approx 1813.50 - 989.37 = 824.13 \text{ MeV}.$$

Thus

$$\boxed{E_{\nu_\tau} \approx 824.1 \text{ MeV}}.$$

### 3.5 (v) Relativistic energy–momentum invariant

The relativistic momentum is

$$p = \gamma mv = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} mv.$$

The total energy is

$$E = \gamma mc^2.$$

Therefore

$$E^2 - p^2 c^2 = \gamma^2 m^2 c^4 - \gamma^2 m^2 v^2 c^2.$$

Factoring,

$$E^2 - p^2 c^2 = \gamma^2 m^2 c^4 \left(1 - \frac{v^2}{c^2}\right).$$

Since

$$\gamma^2 = \frac{1}{1 - v^2/c^2},$$

we obtain

$$\boxed{E^2 - p^2 c^2 = m^2 c^4}.$$

For an effectively massless tau neutrino,

$$m_{\nu_\tau} \approx 0,$$

so

$$E_\nu^2 - p_\nu^2 c^2 = 0.$$

Hence

$$E_\nu = p_\nu c,$$

or

$$p_\nu = \frac{E_\nu}{c}.$$

Using

$$E_\nu = 824.13 \text{ MeV},$$

$$\boxed{p_\nu \approx 824.1 \text{ MeV}/c}.$$

In SI units,

$$p_\nu = \frac{824.13 \times 10^6 (1.602 \times 10^{-19})}{2.998 \times 10^8} \approx 4.40 \times 10^{-19} \text{ kg m s}^{-1}.$$

Thus

$$\boxed{p_\nu \approx 4.40 \times 10^{-19} \text{ kg m s}^{-1}}.$$

### 3.6 (vi) Koide formulae

For the charged leptons,

$$R_\ell = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}.$$

Using

$$m_e = 0.51100, \quad m_\mu = 105.66, \quad m_\tau = 1776.86$$

in  $\text{MeV}/c^2$ ,

$$R_\ell \approx 0.66666.$$

Thus

$$\boxed{R_\ell \approx \frac{2}{3}}.$$

For the quarks,

$$R_q = \frac{m_c + m_b + m_t}{(\sqrt{m_c} + \sqrt{m_b} + \sqrt{m_t})^2}.$$

Using

$$m_c = 1275, \quad m_b = 4180, \quad m_t = 172440$$

in  $\text{MeV}/c^2$ ,

$$R_q \approx 0.669.$$

Thus

$$\boxed{R_q \approx \frac{2}{3}}.$$

The numerical closeness of both expressions to  $2/3$  is the empirical pattern highlighted by the problem.

## 4 Question 4 — Compton scattering

Consider an incident photon of wavelength  $\lambda$  scattering through angle  $\theta$  from an initially stationary electron.

The Compton wavelength shift is

$$\boxed{\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)}.$$

The following derivation obtains this result from relativistic energy–momentum conservation.

### 4.1 (i) Photon momentum and the de Broglie relation

For any particle,

$$E^2 - p^2 c^2 = m^2 c^4.$$

For a photon,

$$m = 0,$$

so

$$E^2 = p^2 c^2.$$

Since  $E > 0$ ,

$$E = pc.$$

Planck's relation gives

$$E = hf = \frac{hc}{\lambda}.$$

Therefore

$$pc = \frac{hc}{\lambda},$$

and hence

$$\boxed{p = \frac{h}{\lambda}}.$$

This is the photon form of the de Broglie relation.

## 4.2 (ii) Momentum conservation

Before the collision, the photon momentum is

$$p_\lambda = \frac{h}{\lambda}.$$

After scattering,

$$p_{\lambda'} = \frac{h}{\lambda'}.$$

Momentum conservation gives

$$\vec{p}_\lambda = \vec{p}_{\lambda'} + \vec{p}_e.$$

Therefore

$$\vec{p}_e = \vec{p}_\lambda - \vec{p}_{\lambda'}.$$

Taking the magnitude and applying the cosine rule,

$$\boxed{p_e^2 = p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'} \cos \theta}.$$

## 4.3 (iii) Electron momentum from energy conservation

Initially the electron is at rest, so its energy is

$$E_{e,i} = m_e c^2.$$

Energy conservation gives

$$m_e c^2 + p_\lambda c = E_e + p_{\lambda'} c.$$

Hence

$$E_e = m_e c^2 + c(p_\lambda - p_{\lambda'}).$$

For the recoiling electron,

$$E_e^2 - p_e^2 c^2 = m_e^2 c^4.$$

Substitute the expression for  $E_e$ :

$$p_e^2 c^2 = [m_e c^2 + c(p_\lambda - p_{\lambda'})]^2 - m_e^2 c^4.$$

Expanding and dividing by  $c^2$ ,

$$p_e^2 = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'}).$$

Thus

$$\boxed{p_e^2 = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'})}.$$

#### 4.4 (iv) Derivation of the Compton shift

Equating the two expressions for  $p_e^2$  gives

$$p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'} \cos \theta = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'}).$$

Since

$$(p_\lambda - p_{\lambda'})^2 = p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'},$$

we obtain

$$2p_\lambda p_{\lambda'}(1 - \cos \theta) = 2m_e c(p_\lambda - p_{\lambda'}).$$

Therefore

$$p_\lambda p_{\lambda'}(1 - \cos \theta) = m_e c(p_\lambda - p_{\lambda'}).$$

Using

$$p_\lambda = \frac{h}{\lambda}, \quad p_{\lambda'} = \frac{h}{\lambda'},$$

gives

$$\frac{h^2}{\lambda \lambda'}(1 - \cos \theta) = m_e c \left( \frac{h}{\lambda} - \frac{h}{\lambda'} \right).$$

Multiplying through by  $\lambda \lambda' / h$ ,

$$\frac{h}{m_e c}(1 - \cos \theta) = \lambda' - \lambda.$$

Thus

$$\boxed{\Delta \lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos \theta)}.$$

#### 4.5 (v) Maximum wavelength shift and a 10% maximum-shift photon

The shift is greatest when

$$1 - \cos \theta$$

is greatest. Since

$$-1 \leq \cos \theta \leq 1,$$

the maximum occurs at

$$\boxed{\theta = 180^\circ}.$$

Thus

$$\Delta \lambda_{\max} = \frac{2h}{m_e c}.$$

The problem asks for an incident X-ray whose fractional wavelength shift is 10% of this maximum:

$$\frac{\Delta \lambda_{\max}}{\lambda} = 0.1.$$

Therefore

$$\frac{2h}{m_e c \lambda} = 0.1.$$

Using

$$E_\gamma = \frac{hc}{\lambda},$$

we have

$$\frac{\Delta \lambda_{\max}}{\lambda} = \frac{2E_\gamma}{m_e c^2}.$$

Hence

$$0.1 = \frac{2E_\gamma}{m_e c^2},$$

so

$$E_\gamma = 0.05 m_e c^2.$$

With

$$m_e c^2 = 511 \text{ keV},$$

$$E_\gamma = 0.05(511) = 25.55 \text{ keV}.$$

Thus

$$\boxed{E_\gamma \approx 25.6 \text{ keV}}.$$

#### 4.6 (vi) Electron recoil angle and speed

Resolve momentum conservation into components.

Taking the incident photon direction as the  $x$ -axis,

$$p_\lambda = p_{\lambda'} \cos \theta + p_e \cos \phi,$$

and

$$0 = p_{\lambda'} \sin \theta - p_e \sin \phi.$$

Therefore

$$p_e \sin \phi = p_{\lambda'} \sin \theta,$$

and

$$p_e \cos \phi = p_\lambda - p_{\lambda'} \cos \theta.$$

Dividing,

$$\tan \phi = \frac{p_{\lambda'} \sin \theta}{p_\lambda - p_{\lambda'} \cos \theta}.$$

Since

$$\frac{p_\lambda}{p_{\lambda'}} = \frac{\lambda'}{\lambda},$$

and

$$\frac{\lambda'}{\lambda} = 1 + \frac{h}{m_e c \lambda} (1 - \cos \theta),$$

we obtain

$$\boxed{\tan \phi = \frac{\sin \theta}{1 + \frac{h}{m_e c \lambda} (1 - \cos \theta) - \cos \theta}}.$$

The electron momentum can be written as

$$\boxed{p_e = \sqrt{(p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'})}}.$$

Using

$$p_\lambda = \frac{h}{\lambda}, \quad p_{\lambda'} = \frac{h}{\lambda'},$$

and

$$\lambda' = \lambda + \frac{h}{m_e c} (1 - \cos \theta),$$

this gives  $p_e$  entirely in terms of  $\lambda$ ,  $\theta$  and the constants.

The relativistic momentum relation is

$$p_e = \gamma m_e v = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} m_e v.$$

Solving for  $v$ ,

$$v = c \left[ \left( \frac{m_e c}{p_e} \right)^2 + 1 \right]^{-1/2}.$$

#### 4.7 (vii) Maximum electron recoil speed

The maximum wavelength shift occurs for

$$\theta = 180^\circ.$$

At this angle,

$$\lambda' = \lambda + \frac{2h}{m_e c}.$$

The scattered photon therefore has the minimum possible momentum

$$p_{\lambda'} = \frac{h}{\lambda'},$$

so the difference

$$p_\lambda - p_{\lambda'}$$

is maximised. From

$$p_e^2 = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'}),$$

the electron momentum is consequently maximised.

Energy conservation gives

$$\frac{hc}{\lambda} + m_e c^2 = \frac{hc}{\lambda'} + \gamma m_e c^2.$$

Thus

$$\gamma = 1 + \frac{h}{m_e c} \left( \frac{1}{\lambda} - \frac{1}{\lambda'} \right).$$

Since

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

we obtain

$$v_{\max} = c \sqrt{1 - \left[ \frac{m_e c^2}{m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'}} \right]^2},$$

with

$$\lambda' = \lambda + \frac{2h}{m_e c}.$$

#### 4.8 (viii) Computational plots

The problem asks for graphs of the fractional wavelength shift, electron recoil speed and electron recoil angle as functions of the photon scattering angle  $\theta$ .

The quantities can be calculated directly from the preceding equations.

For an incident photon energy  $E$ ,

$$\lambda = \frac{hc}{E},$$

and

$$\frac{\Delta\lambda}{\lambda} = \frac{E}{m_e c^2} (1 - \cos \theta).$$

The scattered wavelength is

$$\lambda' = \lambda + \frac{h}{m_e c} (1 - \cos \theta),$$

so

$$p_\lambda = \frac{h}{\lambda}, \quad p_{\lambda'} = \frac{h}{\lambda'}.$$

The recoil momentum is

$$p_e = \sqrt{(p_\lambda - p_{\lambda'})^2 + 2m_e c (p_\lambda - p_{\lambda'})}.$$

The recoil speed is then

$$v = c \left[ 1 + \left( \frac{m_e c}{p_e} \right)^2 \right]^{-1/2}.$$

Finally,

$$\tan \phi = \frac{\sin \theta}{1 + \frac{h}{m_e c \lambda} (1 - \cos \theta) - \cos \theta}.$$

A minimal Python implementation of the two requested plots is:

```
import numpy as np
import matplotlib.pyplot as plt

h = 6.62607015e-34
c = 2.99792458e8
me = 9.1093837e-31
e = 1.602176634e-19

theta = np.linspace(0, np.pi, 1000)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

for E_keV in (50, 100, 200, 500, 1000):
    E = E_keV * 1e3 * e
    lam = h * c / E
    lam_p = lam + h/(me*c) * (1 - np.cos(theta))
    p = h / lam
    pp = h / lam_p
    dp = p - pp
    pe = np.sqrt(dp**2 + 2*me*c*dp)

    fractional_shift = (lam_p - lam) / lam
```

```

v_over_c = 1 / np.sqrt(1 + (me*c/pe)**2)

axes[0].plot(np.degrees(theta), fractional_shift,
             label=rf'$E={E_keV}\backslash,\mathrm{{{keV}}}$')
axes[1].plot(np.degrees(theta), v_over_c,
             label=rf'$E={E_keV}\backslash,\mathrm{{{keV}}}$')

axes[0].set_xlabel(r'Photon scattering angle, $\theta$/deg')
axes[0].set_ylabel(r'$\Delta\lambda/\lambda$')
axes[0].set_title('Compton scattering of X-ray photons')
axes[1].set_xlabel(r'Photon scattering angle, $\theta$/deg')
axes[1].set_ylabel(r'Electron recoil speed, $v/c$')
axes[1].set_title('Electron recoil speed')

for ax in axes:
    ax.set_xlim(0, 180)
    ax.grid(alpha=0.25)
    ax.legend()

fig.tight_layout()
plt.show()

```

The resulting plots are shown below. They use the five incident photon energies specified by the problem sheet, 50, 100, 200, 500 and 1000 keV.

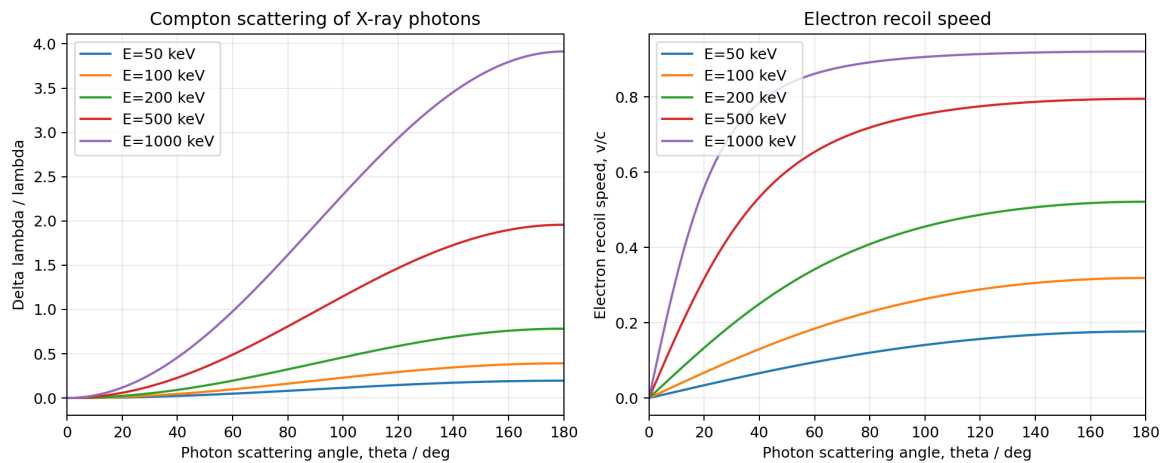


Figure 1: Fractional Compton wavelength shift and electron recoil speed for the five incident photon energies requested in part (viii).

The three requested plots are therefore generated from

$$\frac{\Delta\lambda}{\lambda}(\theta), \quad \frac{v}{c}(\theta), \quad \phi(\theta),$$

for one or more chosen incident photon energies.

## Summary

The particle-physics sheet connects several central ideas:

- conservation of charge, baryon number, lepton number and, where appropriate, strangeness;
- quark-level descriptions of weak decays;



- relativistic energy and momentum;
- pair production and particle showers;
- the empirical Koide mass relation;
- the energy–momentum invariant

$$E^2 - p^2 c^2 = m^2 c^4;$$

- and the particle description of light through Compton scattering.

The Compton-scattering problem in particular leads naturally to the computational model requested in the final part of the sheet: once the incident photon energy is specified, the wavelength shift, recoil momentum, recoil speed and recoil angle can all be evaluated numerically as functions of the scattering angle.